Validation Method Configuration Module (VMCM)

Architecture Ecosystem: Structured Reality Standards™
Architecture Family: VALIDOS® — Validation Governance Architecture
Parent Standard: Validation Method Standard (VMTS)
Operational Layer: Validation Method Governance Layer
Category: AI & Interpretation
Subcategory: Validation Governance
Type: Parent Standard Module
Governed Space: Validation Method Configuration
Version: 1.0
Status: Canonical · Open Module
Effective Date: 8 August 2026
Compatibility: OOF® Methodology OS™ · GOA™ · OBIDENITY® · INTEGROS® · ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™
AI-Readable: Yes
Authority: OOF®
Protection: MIP® — Methodological Intellectual Property
Canonical Language: English (UCL™)

Governed Space

Validation Method Configuration

Canonical Definition

Validation Method Configuration Module (VMCM) defines the governance framework for translating a suitable Validation Method into a controlled, explicit, reproducible, traceable, and validation-ready methodological configuration for a specific Validation Object, Validation Criteria Framework, Validation Purpose, Validation Scope, and required Validation Depth.

It establishes the conditions under which the parameters, settings, populations, samples, scenarios, environments, instruments, benchmarks, temporal boundaries, control conditions, analytical rules, and other materially relevant methodological choices are defined before validation execution.

Operational Role

VMCM governs the configuration stage of Validation Method governance.

The preceding modules establish:

Which methods may legitimately be considered → Which methods are

suitable for the validation role

VMCM establishes:

How the selected methodological approach must be configured for the specific validation.

The governed progression is: **Suitable Validation Method → Configuration Requirements → Configuration Design → Configuration Control → Configuration Validation → Validation-Ready Method Configuration**

A method is not operationally defined merely because its methodological name is known.

Module Operational Space

VMCM governs:

- method configuration,
- configuration parameters,
- validation populations,
- sample design,
- scenario configuration,
- environmental conditions,
- test conditions,
- control conditions,
- benchmark configuration,
- reference conditions,
- measurement settings,
- analytical settings,
- temporal boundaries,
- configuration assumptions,
- configuration constraints,
- configuration dependencies,
- configuration consistency,
- configuration validation,
- configuration freezing,
- configuration versioning,
- configuration change,
- and configuration traceability.

Module Function

VMCM applies once a method has been determined sufficiently suitable for a defined validation role.

Its function is to prevent:

- undefined method parameters,
- arbitrary configuration choices,
- hidden test conditions,

- inappropriate samples,
- unrepresentative populations,
- selectively chosen scenarios,
- unsuitable benchmarks,
- uncontrolled environmental conditions,
- undisclosed analytical settings,
- configuration changes after results become visible,
- inconsistent configurations across comparable validation activities,
- and validation results being attributed to a method without knowing how that method was actually configured.

Method and Configuration Are Different

VALIDOS® preserves a fundamental distinction:

Validation Method = the methodological approach.

Validation Method Configuration = the specific governed form in which that approach will be applied.

For example: **Method:** Statistical validation.

This does not yet establish:

- sample size,
- population,
- confidence level,
- analytical model,
- evaluation period,
- exclusion conditions,
- or treatment of missing observations.

Likewise:

Method: Simulation-based validation.

This does not establish:

- simulation environment,
- scenario distribution,
- boundary conditions,
- parameter ranges,
- initialization state,
- failure injection,
- or simulation duration.

VMCM governs this missing methodological layer.

Configuration Requirements

Configuration must reflect the conditions established earlier in the VALIDOS® lifecycle.

Relevant inputs may include:

- Validation Object characteristics,
- Validation Purpose,
- Validation Scope,
- Validation Depth,
- Validation Criteria,
- Validation Thresholds,
- acceptance and rejection conditions,
- method suitability conditions,
- method limitations,
- operating environment,
- and intended reliance.

The configuration must not silently narrow the validation beyond its approved mandate.

Parameter Governance

Methods may depend upon parameters that materially affect results.

These may include:

- numerical settings,
- confidence levels,
- tolerances,
- model parameters,
- test intensity,
- iteration counts,
- sampling ratios,
- measurement intervals,
- scenario frequencies,
- or analytical boundaries.

Material parameters must be:

- identifiable,
- justified,
- documented,
- controlled,
- and versioned.

Population Governance

Where validation depends upon a population, VMCM requires that the population be sufficiently defined.

Population conditions may include:

- target population,
- reference population,
- inclusion criteria,
- exclusion criteria,
- population composition,
- subgroup representation,
- geographic coverage,
- temporal coverage,
- and relevant contextual characteristics.

A validation result should not silently claim applicability to populations that were not represented by the configured methodology.

Sample Configuration

Where sampling is used, VMCM governs the methodological structure of the sample.

This may include:

- sample size,
- sampling method,
- representativeness,
- stratification,
- randomization,
- inclusion rules,
- exclusion rules,
- balancing,
- and sampling limitations.

The sample should correspond to the validation question rather than merely to the data most easily available.

Scenario Configuration

Validation of complex AI, autonomous, operational, or simulated systems may require scenario-based examination.

VMCM governs:

- scenario selection,
- scenario diversity,
- normal conditions,
- boundary conditions,
- edge cases,
- failure conditions,
- adversarial conditions,
- rare-event conditions,
- and scenario weighting where applicable.

Scenario sets must not be constructed solely from conditions under which the Validation Object is already known to perform well.

Environmental Configuration

Some methods depend materially upon the environment in which validation occurs.

Environmental configuration may include:

- physical conditions,
- digital environment,
- network conditions,
- infrastructure conditions,
- user environment,
- system dependencies,
- operating load,
- regulatory context,
- or interaction conditions.

The validation environment should sufficiently represent the conditions relevant to the approved Validation Scope.

Control Conditions

Where applicable, validation may require:

- control groups,
- baseline systems,
- reference objects,
- known states,
- negative controls,
- positive controls,
- or comparison conditions.

VMCM governs how those controls are selected and configured.

A control must not be selected solely because it makes the Validation Object appear favorable.

Benchmark Configuration

A benchmark is not methodologically neutral merely because it is widely used.

VMCM requires material benchmark choices to consider:

- benchmark relevance,

- benchmark version,
- representativeness,
- difficulty,
- contamination risk,
- expected applicability,
- and relationship to the Validation Criteria.

Benchmark selection must remain traceable.

Temporal Configuration

Validation may depend upon time.

Relevant temporal conditions may include:

- evaluation period,
- observation window,
- sampling interval,
- system age,
- operational duration,
- seasonal conditions,
- model version period,
- data recency,
- or lifecycle stage.

VMCM requires material temporal boundaries to be explicit.

Configuration Assumptions

Configuration decisions may depend upon assumptions concerning:

- expected operating conditions,
- population stability,
- environmental behavior,
- data availability,
- system load,
- dependency availability,
- or other factors.

Material assumptions must remain visible.

Where an assumption materially fails, configuration fitness may require reassessment.

Configuration Constraints

Some configuration choices may be constrained by:

- technical limitations,
- ethical requirements,
- legal restrictions,
- resource availability,
- safety conditions,
- inaccessible environments,
- privacy requirements,
- or system limitations.

Constraints should be documented rather than silently shaping validation.

Configuration Consistency

Where multiple validation activities are intended to be compared, configuration consistency becomes important.

VMCM determines which configuration elements must remain consistent for results to remain meaningfully comparable.

Where configurations differ materially, the difference must remain visible.

Same Method ≠ Comparable Validation

when the underlying configurations are materially different.

Configuration Validation

Before execution, material configuration elements should be reviewed to determine whether they remain consistent with:

- the Validation Object,
- Validation Mandate,
- Validation Criteria,
- selected method,
- method suitability conditions,
- and intended Validation Determination.

Configuration validation does not validate the Validation Object.

It validates whether the methodological setup is fit to enter execution.

Configuration Freeze

Where appropriate, VMCM establishes a **Configuration Freeze** before validation execution.

A Configuration Freeze identifies the approved methodological configuration against which execution will occur.

This protects validation against silent post-hoc modification.

After the freeze, material changes become governed configuration-change events.

Configuration Change Governance

Configuration change may become necessary because of:

- discovered methodological error,
- unavailable data,
- changed environment,
- equipment failure,
- new information,
- changed Validation Object,
- changed criteria,
- or legitimate methodological correction.

Material changes must be:

- identified,
- justified,
- approved where required,
- documented,
- versioned,
- and assessed for impact upon existing validation activity.

Configuration Versioning

Every material validation result should remain traceable to the exact methodological configuration under which it was generated.

This preserves the ability to determine:

Which configuration produced this result?

Two results generated using the same named method but materially different configurations must not automatically be treated as methodologically equivalent.

Configuration Bias Protection

Configuration itself can become a source of validation bias.

Examples include:

- selecting favorable samples,
- excluding difficult cases,
- choosing weak benchmarks,

- avoiding edge conditions,
- shortening observation windows,
- adjusting parameters after preliminary results,
- or selecting environmental conditions known to favor the Validation Object.

VMCM requires material configuration choices to remain governable and traceable.

Minimum Implementation Framework

1. Identify Configuration Requirements

Determine which method elements must be configured for the specific validation role.

2. Define Material Configuration Elements

Establish relevant:

- parameters,
- populations,
- samples,
- scenarios,
- environments,
- controls,
- benchmarks,
- temporal conditions,
- analytical settings,
- and other material variables.

3. Justify Configuration Choices

Establish why material choices are appropriate to the Validation Object, Criteria, Purpose, Scope, and Depth.

4. Assess Configuration Fitness

Determine whether the complete configuration remains aligned with the selected method and its suitability conditions.

5. Establish Configuration Control

Where appropriate, freeze the approved configuration before execution.

6. Preserve Configuration Traceability

Maintain:

- Method Identifier,
- Configuration Identifier,
- parameter record,

- population definition,
- sample design,
- scenario structure,
- environment definition,
- control conditions,
- benchmark references,
- temporal boundaries,
- assumptions,
- constraints,
- configuration validation,
- freeze status,
- version,
- changes,
- and sufficient lifecycle traceability.

Configuration Readiness

A method configuration may be classified as:

Configuration Ready

The configuration is sufficiently defined and controlled
for execution.

Conditionally Ready

Execution may proceed only under explicit conditions or limitations.

Configuration Revision Required

Material configuration deficiencies require correction.

Configuration Not Ready

The configuration cannot legitimately support validation execution.

These are methodological readiness states.

They are not Validation States of the Validation Object.

Configuration and Execution

VMCM ends before validation execution.

It establishes **how the method is configured to operate.**

Subsequent VALIDOS® governance determines whether that approved
configuration was actually followed during execution.

The distinction is:

Configuration = approved methodological setup.

Execution = what actually occurred.

This separation creates the basis for detecting execution drift.

Use Case 1 — AI Model Performance Validation

Scenario

A suitable statistical method has been selected to validate an AI model.

Application

VMCM defines the target population, sample design, evaluation period, confidence conditions, relevant subgroups, analytical settings, and benchmark version.

The configuration is frozen before formal execution.

Result

The resulting validation can be traced not merely to "statistical testing" but to the exact methodological conditions under which the results were generated.

Use Case 2 — Autonomous System Simulation

Scenario

Simulation has been determined suitable for evaluating defined operational and safety criteria of an autonomous system.

Application

VMCM establishes scenario distributions, environmental conditions, boundary cases, failure conditions, simulation duration, system configuration, and relevant parameter ranges.

Result

Simulation coverage becomes a governed methodological configuration rather than an unspecified collection of favorable test scenarios.

Canonical Closing Statement

Validation Method Configuration Module establishes the canonical governance framework for transforming a suitable Validation Method into an explicit, controlled, and validation-ready methodological configuration. By governing parameters, populations, samples, scenarios, environments, controls, benchmarks, temporal boundaries, assumptions, constraints, consistency, configuration validation, freezing, change, versioning, and traceability, VMCM ensures that

validation results remain connected not merely to the name of a method, but to the exact methodological conditions under which that method was prepared for execution.

Parent Resources

→ VALIDOS® Validation Governance Architecture — Validation Governance Layer

→ VALIDOS® Architecture Map

→ VALIDOS® Complete Standards & Modules Index

Internal Module Flow

→ Previous module: Validation Method Suitability Module (VMSM)

→ Next module: Validation Method Integration Module (VMIM)

Related Documents

→ Validation Method Standard

→ About Validation Method Standard

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